Social media analysis

1. Project Overview

This project focuses on analyzing a social media dataset to understand user engagement patterns and interactions across different platforms. The main objective of the analysis is to explore the dataset, clean and validate the data, and apply statistical and visualization techniques to extract meaningful insights.

Using Python data analysis libraries such as pandas, NumPy, matplotlib, and seaborn, the project examines different metrics related to social media activity, including likes, comments, shares, and other engagement indicators. The goal is to identify trends and relationships between these metrics to understand what factors influence user engagement.

The results of this analysis are presented through statistical summaries and visualizations, which help highlight patterns in user behavior and content performance. These insights can be useful for social media managers, marketers, and analysts to improve engagement strategies and optimize content performance.

2. Setup Instructions

Follow these steps to install the required tools and run the project:

1. Install Python (version 3.8 or higher) on your system.
2. Install Anaconda or Jupyter Notebook to run the analysis environment.
3. Install the required Python libraries using the following command in the terminal or Anaconda prompt:

pip install pandas numpy matplotlib seaborn jupyter

4. Download the dataset file Social_Media.csv.

5. Place the dataset in the project directory or in the Downloads folder.
6. Open Jupyter Notebook and navigate to the project folder.
7. Open the analysis notebook file.
8. Load the dataset using the pandas read_csv() function.
9. Run the notebook cells sequentially to perform the analysis and generate visualizations.

3. Code Structure

The project code is organized in a structured manner to maintain readability and maintainability.

```
social-media-analysis/
|
 ├── data/
|   └── Social_Media.csv
|
 ├── notebooks/
|   └── social_media_analysis.ipynb
|
 ├── reports/
|   └── analysis_report.pdf
|
 ├── src/
|   └── data_analysis.py
```

```
│
└── README.md
```

Folder Description

data/
Contains the raw dataset used for analysis.

notebooks/
Contains the Jupyter Notebook used for performing data analysis and generating visualizations.

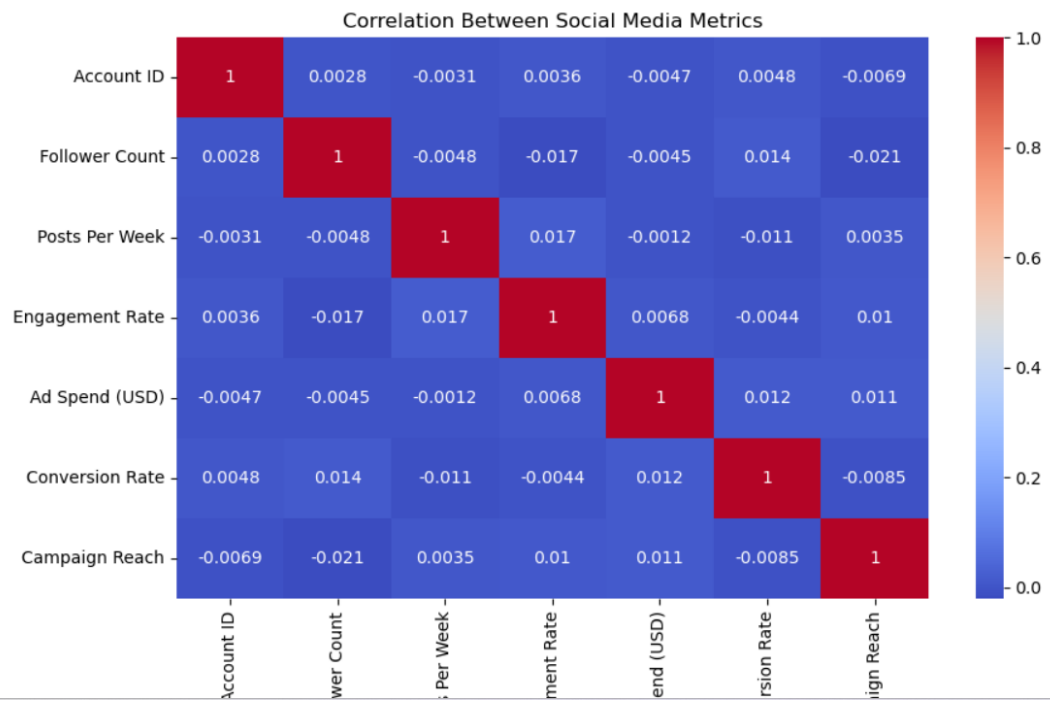
reports/
Contains the final report generated from the analysis, including insights and visualizations.

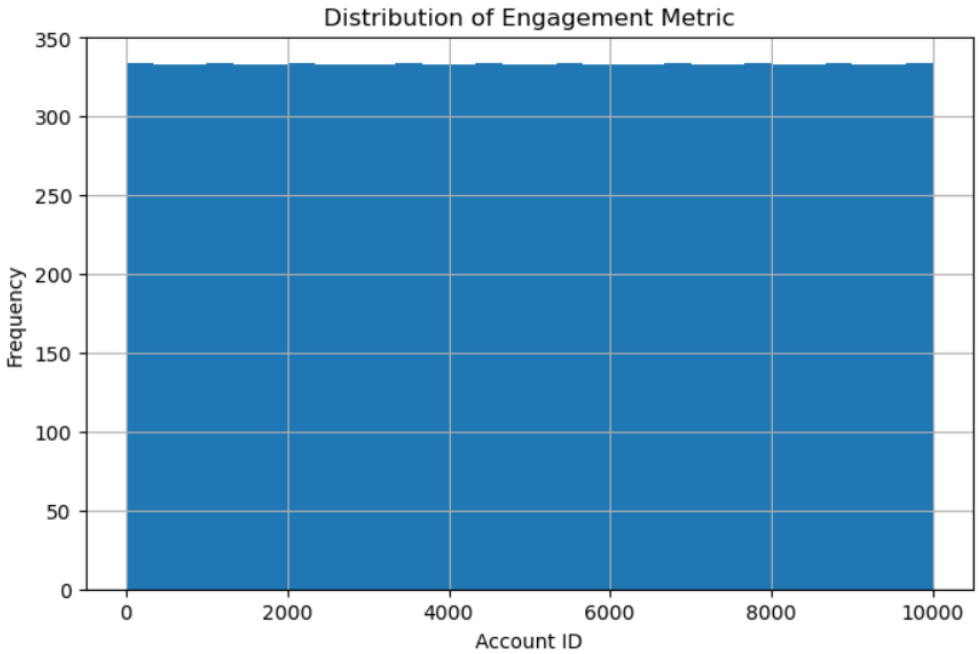
src/
Contains Python scripts used for implementing data processing and analysis functions.

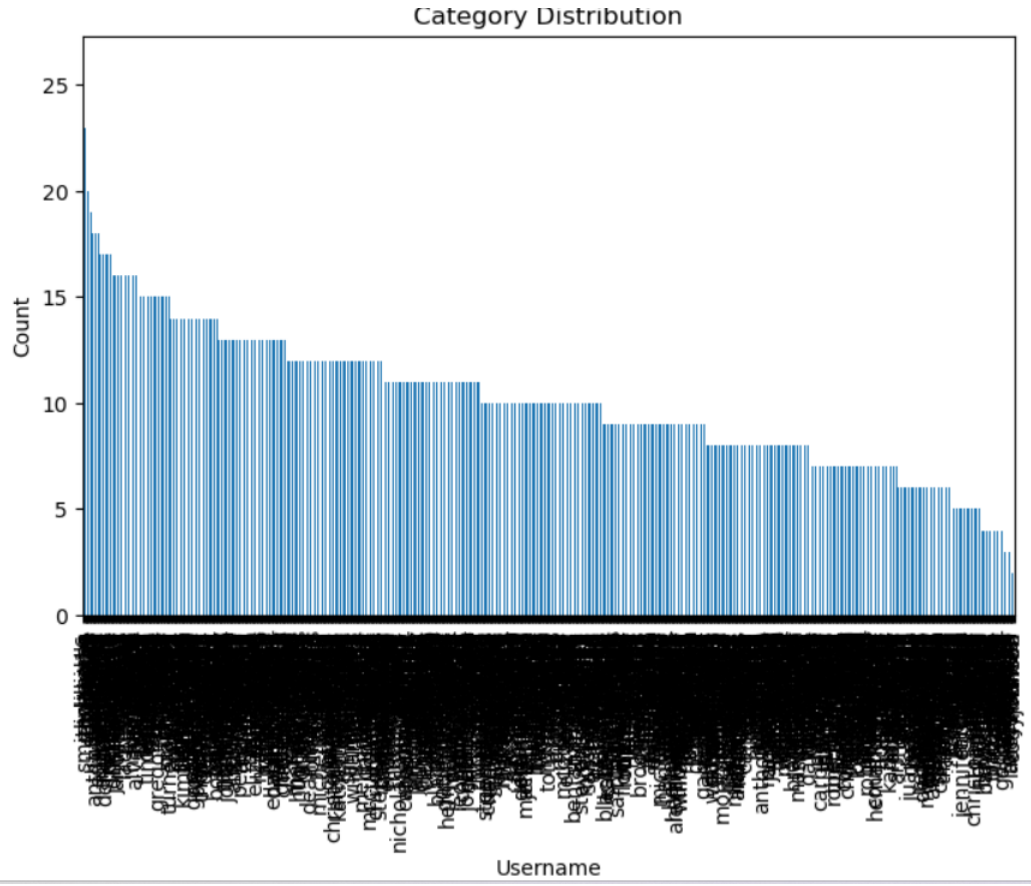
README.md
Provides documentation about the project, setup instructions, and project overview.

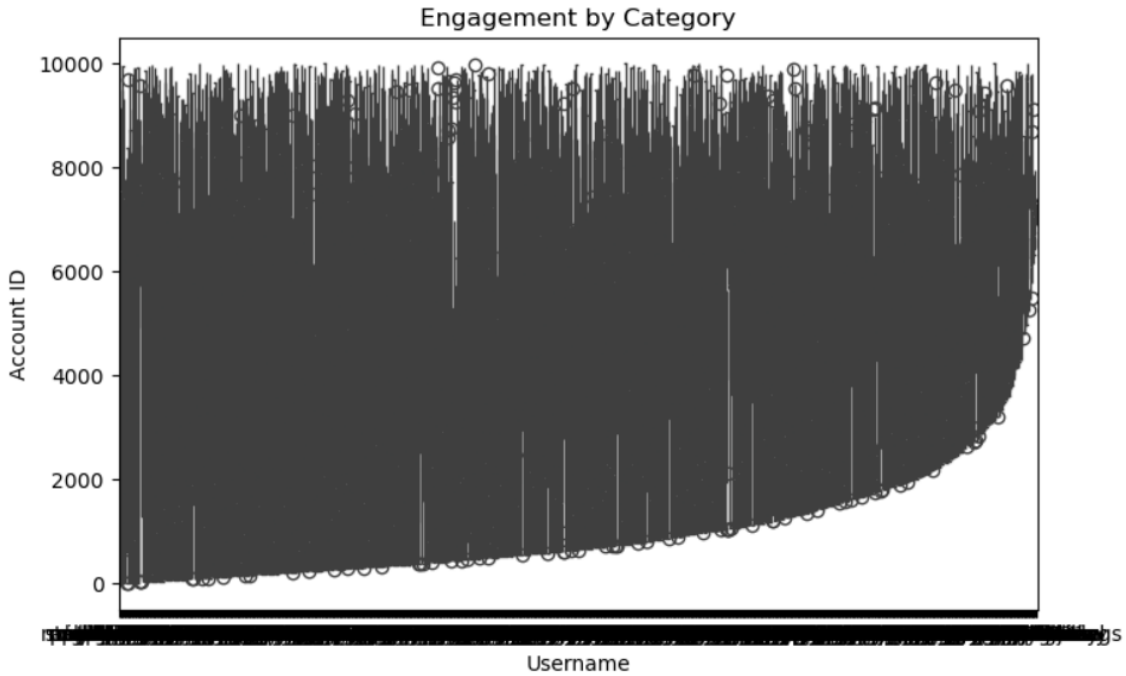
This organized structure ensures that the project is easy to understand, maintain, and share through platforms such as GitHub.

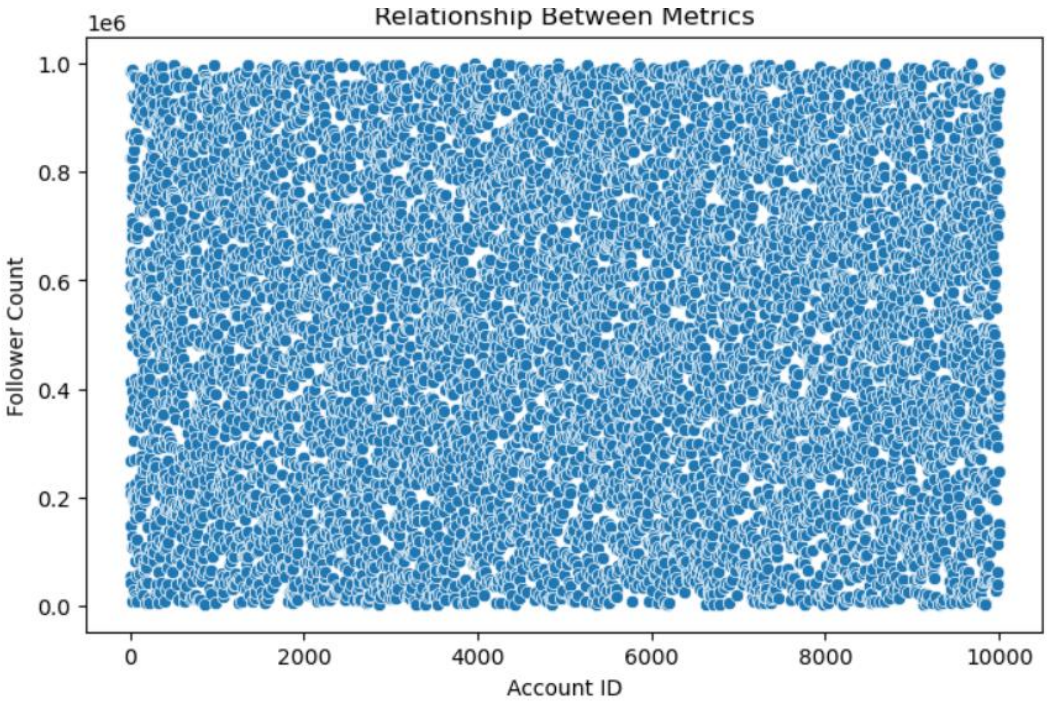
4. **Visualization**


Correlation Between Social Media Metrics
Account ID
Follower Count
Posts Per Week
Engagement Rate
Ad Spend (USD)
Conversion Rate
Campaign Reach
1 0.0028 -0.0031 0.0036 -0.0047 0.0048 -0.0069
0.0028 1 -0.0048 -0.017 -0.0045 0.014 -0.021
-0.0031 -0.0048 1 0.017 -0.0012 -0.011 0.0035
0.0036 -0.017 0.017 1 0.0068 -0.0044 0.01
-0.0047 -0.0045 -0.0012 0.0068 1 0.012 0.011
0.0048 0.014 -0.011 -0.0044 0.012 1 -0.0085
-0.0069 -0.021 0.0035 0.01 0.011 -0.0085 1
1.0
0.8
0.6
0.4
0.2
0.0


Distribution of Engagement Metric
Frequency
350
300
250
200
150
100
50
0
0
2000
4000
6000
8000
10000
Account ID


Category Distribution
Count
25
20
15
10
5
0
Username


Engagement by Category
Account ID
10000
8000
6000
4000
2000
0
Username



5. Technical Details

This project uses Python-based data analysis techniques to process and analyze the social media dataset. The implementation is carried out using widely used data science libraries such as pandas, NumPy, matplotlib, and seaborn.

Algorithms and Methods

Several statistical and analytical techniques are used to understand patterns in the social media data:

- Descriptive Statistics – Measures such as mean, median, minimum, and maximum values are calculated to summarize engagement metrics like likes, comments, and shares.
- Correlation Analysis – A correlation matrix is generated to identify relationships between different engagement metrics and determine how they influence each other.
- Data Aggregation – The groupby() function in pandas is used to analyze engagement trends across categories such as platform type or user group.
- Visualization Techniques – Graphs such as histograms, bar charts, scatter plots, and heatmaps are used to visually represent the distribution and relationships between variables.

These analytical methods help identify patterns in social media engagement and provide meaningful insights.

Data Structures

The project primarily uses pandas DataFrames, which are tabular data structures used to store and manipulate structured data. Each row represents an individual social media record, while each column represents a specific attribute such as likes, comments, or platform.

Additionally, NumPy arrays are used internally for performing efficient numerical operations and statistical computations.

Architecture

The project follows a data analysis pipeline architecture consisting of the following stages:

1. Data Collection – Importing the social media dataset from a CSV file.
2. Data Preprocessing – Cleaning the dataset by handling missing values and removing duplicate records.
3. Data Analysis – Applying statistical calculations to understand engagement patterns.
4. Data Visualization – Generating visual graphs to interpret the results more effectively.
5. Insight Generation – Interpreting the findings to derive meaningful conclusions about social media performance.

This modular workflow ensures that the analysis process is structured, scalable, and easy to maintain.

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6. Testing Evidence

Testing was performed to ensure that the dataset was correctly processed and that the analysis results were reliable.

Data Validation Tests

Several validation checks were applied during data preprocessing:

- Missing Value Check – The isnull().sum() function was used to detect missing values in the dataset.
- Duplicate Record Check – Duplicate entries were identified using the duplicated() function and removed to maintain data integrity.
- Data Type Verification – The info() function was used to verify that columns contained the appropriate data types required for analysis.

Example Test Cases

Test Case 1 – Dataset Loading

Test Case	Expected Result	Status
Load CSV dataset using read_csv()	Dataset loads successfully without errors	Passed

Test Case 2 – Missing Value Detection

Test Case	Expected Result	Status
Run df.isnull().sum()	Missing values are identified correctly	Passed

Test Case 3 – Duplicate Removal

Test Case	Expected Result	Status
Run df.duplicated().sum()	Duplicate rows detected and removed	Passed

Output Validation

The outputs generated from statistical calculations and visualizations were reviewed to confirm correctness. Visualizations such as histograms, bar charts, scatter plots, and heatmaps were checked to ensure they accurately represent the dataset.

Running the Jupyter Notebook cells sequentially verified that the entire workflow executes without errors and produces consistent results.